Benjamin Nie

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Education

University of Maryland - College Park College Park, MD
Bachelor of Science in Computer Science - Machine Learning *May. 2029*

- Activities: Software Engineer, UMD App Dev Club

Experience

S.E.E. Program Teaching Assistant Jun. 2025 - Jul. 2025
Johns Hopkins University *Frederick, MD*

- Guided 20+ students daily in labs/coursework/projects and graded 60 pages of final exams
- Prepared, set up, and maintained 2-3 labs per week for hands-on experiments, ensuring proper calibration and safety

Computational Software Research Intern Aug. 2025 - Present
Mount Saint Mary's University *Frederick, MD*

- Developed a Python pipeline to automate ligand optimization for the BACE1 protein, processing thousands of ligand–receptor atom interactions from AutoDock Vina docking outputs using Pandas and SciPy
- Implemented a greedy pairing algorithm to identify optimal one-to-one ligand–receptor atom pairings by ranking candidates based on spatial distance and receptor partial charge.
- Automated molecular structure optimization by reducing thousands of candidate interactions to 3 target atoms using charge and distance thresholds, then selecting atom replacements based on valence, receptor charge, and molecular composition

Programming Projects Lead Sep. 2023 - Jun. 2026
Urbana High School Future Business Leaders Of America Chapter *Ijamsville, MD*

- Maintained website infrastructure using HTML and CSS for 200+ member FBLA chapter
- CookingMania (2023-2024): Built a multiplayer cooking game using Unity, C#, Audacity, and GitHub; placed 7th at FBLA national competition
- CodeVerse (2024-2025): Built an Android education programming application leveraging Flutter framework and Firebase backend; qualified for and placed 3rd at FBLA state competition

Projects

Scute | *JavaScript, Python, React, Vite, NodeJS, ExpressJS, Flask, PostgreSQL, Render* June 2026 - August 2026

- Built a full-stack campus navigation web application, integrating a custom Python A* pathfinding engine over GraphML map data to generate optimal walking routes
- Developed a secure authentication and user management system with JWT-based authentication, password hashing, input validation, and account recovery
- Improved application performance and reliability by implementing geocoding, interactive route visualization, RESTful APIs, rate limiting, and robust backend error handling
- Deployed frontend, backend, and pathfinding service on Render: https://scute.onrender.com/

Technical Skills

Languages: Java, Python, SQL (Postgres), JavaScript, HTML/CSS
Frameworks: React, Node.js, Flask, ExpressJS
Developer Tools: Git, VS Code, Visual Studio
Certifications: Certiport Information Technology Specialist